Yifei Dong

Lindstedtsvägen 24, 114 28 Stockholm
Sweden
Website: *yifeidong0.github.io/*
Email: *yifeid@kth.se*

Education

2025 **Duke University**, *Visiting Scholar*, Robot Dexterity Lab, with Prof. Xianyi Cheng

2023 - Now **KTH Royal Institute of Technology**, *PhD Candidate*, Robotics, Computer Science

2019 - 2022 **ETH Zurich**, *M.Sc.*, Mechanical Engineering

2015 - 2019 **Shanghai Jiao Tong University**, *B.Sc.*, Mechanical Engineering
Tsien Hsue-shen Program & Zhiyuan Honors Program of Engineering

Employment

2023 - Now **Robotics, Perception and Learning, KTH**, *Stockholm*
Doctoral Student, with Assoc. Prof. Florian Pokorny

2022 **Idiap, EPFL**, *Martigny*, *Research Assistant*, with Dr. Sylvain Calinon
Phase estimation on Riemannian manifold. Code contribution: Robotics Code From Scratch

2022 **F&P Robotics**, *Zürich*, *Software Engineer Intern*
Motion planning of door opening and human-robot interaction for a mobile service robot.

2018 **NIO Inc.**, *Shanghai*, *Software Engineer Intern*
LiDAR simulation of an autonomous vehicle in various traffic scenes.

Publications

Journals and Conference Proceedings (* indicates equal contribution):

[J1] Yifei Dong, Xianyi Cheng, Florian T. Pokorny, *Characterizing Manipulation Robustness through Energy Margin and Caging Analysis*, Robotics and Automation Letters (RA-L), 2024, presented at ICRA 2025, *Paper*

[J2] Haofei Lu, Yifei Dong, Zehang Weng, Florian T. Pokorny, Jens Lundell, Danica Kragic, *Grasping a Handful: Sequential Multi-Object Dexterous Grasp Generation*, Robotics and Automation Letters (RA-L), 2025, *Paper* | *Website*

[C1] Yifei Dong*, Yan Zhang*, Sylvain Calinon, Florian T. Pokorny, *Robustness-Aware Tool Selection and Manipulation Planning with Learned Energy-Informed Guidance*, International Conference on Robotics and Automation (ICRA), 2026, *ArXiv*

[C2] Yifei Dong*, Shaohang Han*, Xianyi Cheng, Werner Friedl, Rafael I. Cabral Muchacho, Máximo A. Roa, Jana Tumova, and Florian T. Pokorny, *CageCoOpt: Enhancing Manipulation Robustness through Caging-Guided Morphology and Policy Co-Optimization*, International Conference on Intelligent Robots and Systems (IROS), 2025, *Paper* | *Website*

[C3] Yifei Dong, Florian T. Pokorny, *Quasi-static Soft Fixture Analysis of Rigid and Deformable Objects*, International Conference on Robotics and Automation (ICRA), 2024, *Paper* | *Website* | *Workshop*

[C4] David Cáceres Domínguez et al., *The First WARA Robotics Mobile Manipulation Challenge–Lessons Learned*, European Conference on Mobile Robots, 2025, *Paper* | *Video*

Kei Ikemura, Yifei Dong, David Blanco-Mulero, Alberta Longhini, Li Chen, Florian T. Pokorny, *Efficient End-effector Co-Design by Demonstration for Deformable Fragile Object Manipulation*, RSS Workshop on Robot Hardware-Aware Intelligence (Spotlight), 2025, *OpenReview*

Preprints:

[1] David Blanco-Mulero, Yifei Dong, Julia Borras, Florian T. Pokorny and Carme Torras, *T-DOM: A Taxonomy for Robotic Manipulation of Deformable Objects*, under review, *Preprint* | *Website*

[2] Kei Ikemura*, Yifei Dong*, David Blanco-Mulero, Alberta Longhini, Li Chen, Florian T. Pokorny, *Sim-to-Real Gentle Manipulation of Deformable and Fragile Objects with Stress-Guided Reinforcement Learning*, under review, *ArXiv*

[3] Kei Ikemura*, Yifei Dong*, Florian T. Pokorny, *Latent Diffeomorphic Co-Design of End-Effectors for Deformable and Fragile Object Manipulation*, under review, *ArXiv*

[4] Christoffer Koo Øhrstrøm, Rafael I. Cabral Muchacho, Yifei Dong, Filippos Moumtzidellis, Ronja Güldenring, Florian T. Pokorny, Lazaros Nalpantidis, *Where to Attend: A Principled Vision-Centric Position Encoding with Parabolas*, under review, *ArXiv*

Academic Activities

Talk, Academic Service, Supervision and Teaching:

2026 Corresponding organizer of ICRA 2026 Workshop on manipulation robustness, Vienna | *Website*

2025 Guest lecture at Duke University's graduate course, Robotic Manipulation, 2025 Spring

2024 Invited talk at Prof. Joel Burdick's group, California Institute of Technology

2024 - Now Reviewer for RA-L, ICRA, IROS, ISRR

2024 - 2026 Supervised master students, Kei Ikemura, Li Chen, Gawtam Chithra Ramesh, Zhenyuan Liang, Hongjia Liu

2024, 2025 Teaching assistant in Introduction to Robotics, DD2410, KTH

Experiences

2024 **Mobile Manipulation Challenge, ABB**, *Västerås, Sweden*
Team member of KTH: Custom gripper co-design for multi-task manipulation | Slides

2021 **Robotic Systems Lab, ETH**, *Zürich*, *Graduate Researcher*, with Prof. Marco Hutter
Master thesis: Mobile door state estimation and parameter identification | Thesis
Semester thesis: Contact-implicit MPC for mobile manipulation | Video | Thesis

2020 **Computer Vision and Geometry Lab & Microsoft**, *Zürich*
Course project: Object mesh reconstruction using RGB-D cameras. | Thesis | Git | Video

2019 **SAIC Motor**, *Shanghai*
Bachelor thesis: Strategy optimization of autonomous emergency braking

2018 **Institute of Design and Control Engineering, SJTU**, *Shanghai*
3-DOF plum processing system for surface-curving and core-deprivation. | Video

2017 **Institute of Intelligent Vehicle, SJTU**, *Shanghai*
Structural design and locomotion formulation of a snake robot. | Video

Awards & Grants

2024 Travel Grant from the Karl Engvers Foundation

2021, 2022 ETH Scholarship for International Students

2019 Excellence Award, BAIC Automobile Invitational Tournament

2018 Tan Kah Kee Invention Award

2016, 2018 General Motors Scholarship

2017 Huawei Scholarship

2016, 2017 Zhiyuan College Honors Scholarship

Skills

Programming Python, C++, MATLAB, C, C#

Tools ROS2, Pybullet, Isaac Gym, Git, Docker, Blender, PyTorch, Simulink, Unity3D, ANSYS, SolidWorks, CATIA, NX Unigraphics

Language English (TOEFL 109/120), Chinese (native), German (Goethe B1)